

Cocycle-Stable Weak-(1, 1) Transference and Spectral Multipliers on Quantum Tori

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Status: candidate partial result, likely valid. Weak-(1, 1) Herz–Schur bounds transfer, without loss of constant, to Fourier multipliers on every cocycle-twisted group von Neumann algebra of a discrete amenable group. Consequently the spectral multiplier theorem of Cadilhac–Conde-Alonso–Parcet yields deformation-uniform complete weak-(1, 1) estimates on all quantum tori. The broader nonconvolution Calderón–Zygmund problem remains open.

1 Source problem and scope

Cadilhac, Conde-Alonso, and Parcet prove weak- L_1 endpoint estimates for matrix-valued Calderón–Zygmund operators and for Mikhlin multipliers in group von Neumann algebras [1]. Their Remark 2.5, on arXiv PDF page 13, asks for a weak endpoint theory in fully noncommutative geometric settings and names quantum Euclidean spaces and the hyperfinite II_1 factor as candidates.

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Remark 2.5. Another open problem related to our work above is to provide Calderón–Zygmund decompositions in fully noncommutative geometric settings and subsequently to establish weak type $(1, 1)$ bounds for some version of singular integrals. Two possible candidates are quantum Euclidean spaces —for which good notions of distance and CZ operators were established in [14]— and the hyperfinite II_1 factor, which is naturally equipped with a dyadic filtration.

The result below treats a substantial convolution subfamily. It applies to arbitrary unitary cocycle twists of discrete amenable groups. In particular, it gives uniform weak endpoints on quantum tori, which are twisted group von Neumann algebras of $\mathbb{Z}^d$. It does not construct the requested Calderón–Zygmund decomposition, cover nonconvolution kernels, or settle the locally compact quantum Euclidean algebra $\mathcal{R}_\Theta$.

2 Definitions

Let Γ be a discrete group and let $\sigma : \Gamma \times \Gamma \rightarrow \mathbb{T}$ be a normalized unitary 2-cocycle. Thus

$$\sigma(g, h)\sigma(gh, k) = \sigma(h, k)\sigma(g, hk), \quad \sigma(e, g) = \sigma(g, e) = 1.$$

The twisted left regular representation on $\ell_2(\Gamma)$ is

$$\lambda_\sigma(g)\delta_h = \sigma(g, h)\delta_{gh}.$$

Its weak operator closure is the twisted group von Neumann algebra $L_\sigma(\Gamma)$. It carries the canonical trace

$$\tau_\sigma\left(\sum_g a_g \lambda_\sigma(g)\right) = a_e.$$

Given $M : \Gamma \rightarrow \mathbb{C}$, define the twisted Fourier multiplier on finite Fourier sums by

$$T_M^\sigma(\lambda_\sigma(g)) = M(g)\lambda_\sigma(g).$$

The corresponding Herz–Schur multiplier on $\mathcal{B}(\ell_2\Gamma)$ is

$$S_M([a_{r,s}]) = [M(rs^{-1})a_{r,s}].$$

For a semifinite tracial von Neumann algebra $(\mathcal{N}, \tau)$, write

$$\|x\|_{1,\infty} = \sup_{t>0} t \tau(\mathbf{1}_{(t,\infty)}(|x|)).$$

3 Idea of the proof

The ordinary amenable transference argument compresses the left regular representation to a Følner net and places the compressions in a tracial ultraproduct. A twist changes each nonzero matrix entry only by a scalar of modulus one. If the (r, s) entry of $\lambda_\sigma(g)$ is nonzero, then $r = gs$, hence $rs^{-1} = g$. Schur multiplication by $M(rs^{-1})$ therefore acts on that entry by exactly $M(g)$, independent of the cocycle.

The only remaining issue is the endpoint topology. Finite compressions obey the assumed Schur weak- $(1, 1)$ estimate. Amenability makes them asymptotically multiplicative in normalized Hilbert–Schmidt norm, producing a trace-preserving embedding of $L_\sigma(\Gamma)$ into their tracial ultraproduct. Trace preservation passes spectral distribution functions at their continuity points, which is enough to recover the weak- L_1 quasi-norm. Thus there is no loss in the endpoint constant.

4 Twisted weak endpoint transference

Theorem 4.1 (Twisted weak- $(1, 1)$ transference). *Let Γ be a discrete amenable group, let σ be a normalized unitary 2-cocycle on Γ , and let $M : \Gamma \rightarrow \mathbb{C}$. Suppose*

$$\|S_M(A)\|_{S_{1,\infty}(\Gamma)} \leq C \|A\|_{S_1(\Gamma)} \quad (A \in S_1(\ell_2\Gamma)).$$

Then T_M^σ extends uniquely to a bounded map

$$T_M^\sigma : L_1(L_\sigma\Gamma) \longrightarrow L_{1,\infty}(L_\sigma\Gamma)$$

with norm at most C .

More generally, if the same Schur estimate holds after every matrix amplification with a common constant C , then every matrix amplification of T_M^σ satisfies the corresponding estimate with the same constant. The bound is independent of σ .

Proof. Choose a left Følner net $(F_i)_i$ in Γ and let P_i be the orthogonal projection of $\ell_2(\Gamma)$ onto $\ell_2(F_i)$. Equip $\mathcal{B}(\ell_2 F_i)$ with normalized trace $\text{tr}_i = |F_i|^{-1} \text{Tr}$. Put

$$j_i(x) = P_i x P_i \quad (x \in L_\sigma\Gamma).$$

For $g, h \in \Gamma$,

$$\begin{aligned} j_i(\lambda_\sigma(g))j_i(\lambda_\sigma(h)) - j_i(\lambda_\sigma(g)\lambda_\sigma(h)) \\ = -P_i\lambda_\sigma(g)(\mathbf{1} - P_i)\lambda_\sigma(h)P_i. \end{aligned}$$

The operator on the right can be nonzero only on basis vectors δ_s with $s \in F_i$ and $hs \notin F_i$. Since all cocycle factors have modulus one,

$$\|j_i(\lambda_\sigma(g))j_i(\lambda_\sigma(h)) - j_i(\lambda_\sigma(g)\lambda_\sigma(h))\|_{2, \text{tr}_i}^2 \leq \frac{|hF_i \triangle F_i|}{|F_i|} \longrightarrow 0.$$

Also

$$\text{tr}_i(j_i(\lambda_\sigma(g))) = \begin{cases} 1, & g = e, \\ 0, & g \neq e. \end{cases}$$

Hence, for a free ultrafilter refining the net, the map

$$J : L_\sigma(\Gamma) \longrightarrow \prod_{\mathcal{U}} (\mathcal{B}(\ell_2 F_i), \text{tr}_i), \quad J(x) = [j_i(x)]_{\mathcal{U}},$$

extends from the twisted group algebra to a trace-preserving $*$ -monomorphism. In particular it preserves L_1 norms and spectral distributions.

We next record the exact intertwining. The matrix coefficients of a twisted group unitary are

$$\langle \lambda_\sigma(g)\delta_s, \delta_r \rangle = \begin{cases} \sigma(g, s), & r = gs, \\ 0, & r \neq gs. \end{cases}$$

Whenever the coefficient is nonzero, $rs^{-1} = g$. Therefore

$$S_M(j_i(x)) = j_i(T_M^\sigma x) \tag{4.1}$$

for every finite twisted Fourier sum x and every i .

Regard a finite corner as an operator on $\ell_2 \Gamma$ by extending it by zero. Normalizing both the L_1 and weak- L_1 quantities by $|F_i|$ does not change the assumed constant. From (4.1), for every $t > 0$,

$$t \text{tr}_i(\mathbf{1}_{(t, \infty)}(|j_i(T_M^\sigma x)|)) \leq C \text{tr}_i(|j_i(x)|). \tag{4.2}$$

The operators in (4.2) are uniformly bounded in operator norm when x is a finite Fourier sum. Continuous functional calculus in the tracial ultraproduct implies

$$\lim_{\mathcal{U}} \text{tr}_i(|j_i(x)|) = \tau_\sigma(|x|)$$

and, at every continuity point t of the spectral distribution of $|T_M^\sigma x|$,

$$\lim_{\mathcal{U}} \text{tr}_i(\mathbf{1}_{(t, \infty)}(|j_i(T_M^\sigma x)|)) = \tau_\sigma(\mathbf{1}_{(t, \infty)}(|T_M^\sigma x|)).$$

Taking the ultralimit in (4.2) gives the desired distribution estimate at all such t . The discontinuity set is countable, and continuity points from above approximate every $t > 0$; consequently their supremum already equals the weak- L_1 quasi-norm. Thus

$$\|T_M^\sigma x\|_{1, \infty} \leq C \|x\|_1$$

for all finite twisted Fourier sums x . Density and completeness of $L_{1, \infty}$ give the unique extension.

For matrix amplification, replace j_i by $\text{id}_{M_k} \otimes j_i$ and use the amplified Schur estimate. Every preceding identity is unchanged. $\square$

5 Quantum-torus consequence

Let $\mathcal{A}''_\Theta$ denote the von Neumann algebra of a d -dimensional quantum torus. It is the twisted group von Neumann algebra $L_{\sigma_\Theta}(\mathbb{Z}^d)$ for the standard cocycle associated with the antisymmetric deformation matrix Θ ; see González-Pérez–Junge–Parcet for the quantum Euclidean and torus framework [2]. Write U_Θ^k for its Fourier unitaries and let

$$\Delta_\Theta(U_\Theta^k) = |k|^2 U_\Theta^k.$$

Corollary 5.1 (Uniform quantum-torus Mikhlin endpoint). *Let $m : \mathbb{R}_+ \rightarrow \mathbb{C}$ satisfy*

$$|m^{(r)}(s)| \leq A_r s^{-r} \quad (s > 0), \quad 0 \leq r \leq \left\lfloor \frac{d}{2} \right\rfloor + 1,$$

and let $m(0)$ be bounded. Then, for every antisymmetric Θ ,

$$m(\Delta_\Theta) : L_1(\mathcal{A}''_\Theta) \longrightarrow L_{1,\infty}(\mathcal{A}''_\Theta)$$

is completely bounded. Its completely amplified weak-(1,1) constant depends only on d , $|m(0)|$, and the displayed Mikhlin constants, and is independent of Θ .

Proof. On $\Gamma = \mathbb{Z}^d$, the conditionally negative length $\psi(k) = |k|^2$ is represented by the d -dimensional cocycle $\beta(k) = k \in \mathbb{R}^d$. Theorem C(i) of Cadilhac–Conde–Alonso–Parcet gives the completely amplified weak-(1,1) estimate for the Herz–Schur multiplier with symbol $m \circ \psi$ under the stated Mikhlin condition [1]. Apply Theorem 4.1 with the arbitrary twist σ_Θ . The value at 0 contributes only the trace-preserving zero-frequency projection and is harmless. $\square$

Remark 5.2. For the standard irrational factor cases, the weak closure of the quantum torus is a hyperfinite II_1 factor. The corollary therefore also gives a concrete singular-multiplier family on the second class named in Remark 2.5. It is not a general weak endpoint theorem for that factor and does not use its dyadic filtration.

6 Verification and novelty status

The proof was audited through the following independent checks.

- The cocycle identity is used only to make λ_σ a projective representation. The intertwining itself uses solely the support relation $r = gs$, so no hidden phase cancellation occurs.
- The Følner boundary estimate is normalized and uniform in the two group elements. It is enough to obtain multiplicativity on finite Fourier sums and then the trace-preserving ultraproduct embedding.
- The weak endpoint is passed at continuity points of the spectral distribution, avoiding an unjustified assertion that spectral projections commute pointwise with arbitrary ultralimits.
- The fixed matrix-amplification argument uses exactly the complete Schur endpoint supplied in Theorem C(i) of the source.

No computational experiment is relevant to the proof. A separate line-by-line report is included as `verification.md`.

A bounded search through 22 July 2026 checked the run registry, the local arXiv corpus, and web/arXiv queries for weak-(1, 1) twisted Fourier transference, cocycle-twisted group von Neumann algebras, quantum-torus weak endpoints, and Mikhlin multipliers. No explicit duplicate was found. Strong L_p twisted transference is known, and a 2025 seminar announcement also describes cocycle-uniform L_p transference, but the search did not locate the weak- L_1 theorem above. Novelty confidence is therefore moderate, not high.

The principal human-review targets are the spectral-distribution passage in Theorem 4.1 and a broader specialist literature check for an implicit endpoint version of twisted transference.

References

- [1] L. Cadilhac, J. M. Conde-Alonso, and J. Parcet, *Spectral multipliers in group algebras and noncommutative Calderón–Zygmund theory*, arXiv:2105.05036, 2021.
- [2] A. M. González-Pérez, M. Junge, and J. Parcet, *Singular integrals in quantum Euclidean spaces*, arXiv:1705.01081, 2017.